Search Search

The Department of Computer Science hosts a computing cluster for scientific
workloads especially development purposes and GPU computing. If you are into bulk number crunching, please also consider the HPC Services offered by ZID [https://www.uibk.ac.at/zid/systeme/hpc-systeme/#HDR3] as they may suit your requirements better.

Getting Access

You have to request an account on the ifi-cluster. Then you can log into the system with your IFI credentials using SSH on **ifi-cluster.uibk.ac.at**. To request access write to ifi-sysadmin@informatik.uibk.ac.at [mailto:ifi-sysadmin@informatik.uibk.ac.at]. The mail should contain your Name, your c-number, your email-address and your research group.

Hardware

The System is located in the IFI server room (3W06) in the ICT Building. It consists of 18 Nodes with the following specification:

```
Motherboard: ASRock Fatal1ty X399 Professional Gaming
CPU: AMD Ryzen Threadripper 2950X WOF
Memory: G.Skill D4128GB 2400-15 Flare X K8
SSD: WD Black SN750 500 GB, NVMe 1.3 , Read: 3470 MB/s, Write: 2600 MB/s
```

Node	CPU	Cores	Main Memory	GPU	GPU Architecture
Headnode	AMD Ryzen 2950X	16	128 GB	1x GeForce RTX 2070 SUPER, 8 GB (GDDR6, 256 Bit)	Turing
gc**1**-gc**7**	AMD Ryzen 2950X	16	128 GB	4x GeForce RTX 2070, 8 GB (GDDR6, 256 Bit)	Turing
gc**8**-gc**16**	AMD Ryzen 2950X	16	256 GB	4x GeForce RTX 2070 SUPER, 8 GB (GDDR6, 256 Bit)	Turing
gc**17**	AMD Ryzen 2950X	16	256 GB	1x NVIDIA TITAN RTX, 24 GB (GDDR6, 384 Bit)	Turing
gc**18**	AMD Ryzen 7985WX	64	512 GB	4x GeForce RTX 4090, 24 GB (GDDR6, 384 Bit)	Ada Lovelace
gc**19**	AMD EPYC 9654	96	1,5TB	8x NVIDIA L40S, 48GB	Ada Lovelace
gc**20**-gc**27**	AMD RYZEN AI MAX+ 395	32	96(GPU)+32(OS)GB	Radeon 8060S	

Connectivity is through a 10G Netgear XS724EM switch

Network

The nodes and storage are interconnected through a 10 Gigabit Ethernet Switch. The upstream connection of the headnode is also 10 Gigabit Ethernet.

System Configuration

- All nodes run Ubuntu Server 24.04 LTS (Jammy Jellyfish).
- Nodes are booted over PXE from headnode and OS is kept in RAM, making them software-wize identical
- All nodes run CUDA Version: 11.8
- All nodes have the same directory layout mounted.
- SLURM is used as job scheduler.
- IFI Auth is used as authentication backend.

Q/A, Caveats

IFI Internal Chat

Subscribe and use the IFI GPU Cluster chat room at Matrix Chat [https://chat.uibk.ac.at/#/room/#ifi-cluster:uibk.ac.at].

ToDos/Changes/Errors

Information about ToDo's, changes and reboot log is collected here here.

Storage

Storage	Path (on all nodes)	Usage
home directory	/home/[username]	for storing important files (mounted from ifi-nas), daily backup
scratch	/scratch/[username]	shared scratchpad storage for big data during computation, might be deleted without explicit notice, no backups
ifi-nas	/mnt/ifi/*[your_group]*	persistent storage, daily backup
ifi-nas	/mnt/ifi-ssd/*[your_group]*	persistent storage, no backup
shared software	/software-shared	if you want to share some software setup with different cluster users
local storage	/mnt/local-disk	local disk space on each node

How to use the storage

Your **home directory** /home/name.surname should not exceed 100GB.Permission should be 700 in case you do not share it.

```
$ chmod 700 /home/name.surname
```

A 18TB HDD space based on BeeGFS is available to the GPU Cluster as a **scratch space**.

Mount point is : `/scratch` on each node

For better usage, please create a directory for yourself using the same username you have for the cluster and set accessibility only to you.

(in case you need to share a directory, give 750 or 755 permissions for group/others)

```
$ cd /scratch
$ mkdir name.surname
$ chmod -R 700 /scratch/name.surname
```

The **IFI-NAS storage** is mounted at `/ifi-NAS/[your_group]`. Information about IFI-NAS is available here.

Adding shared software:

You can create your own directory in the `/software-shared` directory and add/compile your software there. The /software-shared directory is mounted on all nodes

Mounting an external resource on headnode

First create your own directory:

```
mkdir /mnt/users-mount/[your_dir]
```

Then mount the external resource:

```
sudo mount server:/resource /mnt/users-mount/[your_dir]
```

When finished, please do not forget to unmount it and delete the directory:

```
sudo umount /mnt/users-mount/[your_dir]
rm -rf /mnt/users-mount/[your_dir]
```

Installed Software Packages

Ubuntu Distribution Packages from 24.04

Topology

To be linked here soon…

Usage

Show running Jobs

```
squeue
```

Submit Job

Batch Job

Write a configuration file similar to this basic example:

cluster-test.sh

```
#!/bin/bash -l
#SBATCH --partition=IFIgpu
```

```
#SBATCH --job-name=firstDemo
#SBATCH --mail-type=BEGIN,END,FAIL ##optional
#SBATCH --mail-user=max.mustermann@uibk.ac.at ##optional
#SBATCH --account=your_group ##change to your group
#SBATCH --uid=your_username  ##change to your username
#SBATCH --nodes=2
#SBATCH --ntasks-per-node=1
#SBATCH --mem=4G
#SBATCH --gres=gpu:1 ## optional to request specific resources (1 GPU)
#SBATCH --time=0-00:30:00
#SBATCH --output slurm.%N.%j.out # STDOUT
#SBATCH --error slurm.%N.%j.err # STDERR
srun /bin/hostname | /usr/bin/sort
```

Option	Description
--partition=	specifies on which partition you want to run your job. Available partitions are: `IFIall` - for CPU computation, `IFIall` - for GPU computation , `IFItitan` - for large GPU computation on a NVIDIA Titan GPU Card
--job-name=	you can define a name for your job (this field is optional)
--mail-type=	to receive an e-mail if desired. Valid type values are `NONE`, `BEGIN`, `END`, `FAIL`, `REQUEUE`, `ALL`, +others see official documentation (this field is optional)
--mail-user=	specify the users e-mail (it could be any UIBK e-mail). if --mail-type is specified to other then `NONE` and --mail-user is missing e-mails will be send to local user on the IFI cluster (check it with “mail” command on headnode). For system wide mail forwarding please check here
--account=	specify the group you are belonging to
--uid=	specify your_username
--nodes=	define on how many nodes your job should run. The scheduler will allocate the number of nodes to your job.
--ntasks-per-node=	Request that `ntasks` be invoked on each node
--mem=	Specify the real memory required per node - default units are megabytes. Different units can be specified using the suffix [K|M|G|T]
--time=	Set a limit on the total run time of the job allocation. If the requested time limit exceeds the partition's time limit, the job will be left in a PENDING state (possibly indefinitely).
--output=<filename pattern>	The batch script's standard output will be directly to the file name specified in the “filename pattern”. By default both standard output and standard error are directed to the same file.
--error=<filename pattern>	The batch script's standard error will be directed to the file name specified in the “filename pattern”.

The command that you want to run:

```
srun /bin/hostname | /usr/bin/sort
```

Run the file with **`sbatch --test-only <<filename>>`** to check the syntax and with **`sbatch «filename»`** to submit it.

Other valid and useful options:

Option	Description
--exclusive[=user]	The job allocation can not share nodes with other running jobs (or just other users with the “=user” option).
--gres=<list>	Specifies a comma delimited list of generic consumable resources (GPUs). The format of each entry on the list is “name:type:count” (i.e - -gres=gpu:4). You can use this option to request less resources as

Option	Description
	default i.e. - -grep=gpu:1 to allocate only one GPU instead of default 4

There is also an example script for C-Compilation.

Use the official website documentation for details: https://slurm.schedmd.com/sbatch.html [https://slurm.schedmd.com/sbatch.html]

Interactive Job

```
srun /bin/hostname
```

Interactive shell session

```
srun --nodes=1 --nodelist=gc3 --ntasks-per-node=1 --time=01:00:00 --pty bash -i
```

will get you a shell on gc3 node for one hour

Cancel Job

```
scancel <<jobid>>
```

The basic commands can be found in the SLURM User Guide [https://slurm.schedmd.com/quickstart.html]

Using CUDA

"/usr/local/cuda-11.8" and "/usr/local/cuda-11.7" are now installed

Please make sure that you set the correct environment for the version you need:

```
PATH includes /usr/local/cuda-11.x/bin
LD_LIBRARY_PATH includes /usr/local/cuda-11.x/lib64
```

Available partitions

IFIall - nodes - gc[1-19]

IFIgpu2070 - nodes gc1-gc7 ~~- MaxTime=12:00:00 DefaultTime=00:30:00~~

IFIgpu2070S - nodes gc8-gc16

IFItitan - node gc17

IFIgpu4090 - node gc18 - restricted usage. only for group TCS

IFIgpuL40S - node gc19 with 8xL40 GPUs - MaxTime=02:00:00 DefaultTime=00:30:00; please use gres to request only the amount of GPUs you need not all 8!

IFIAMD - nodes gc20-gc27 - restricted usage. only for group DPS, EdgeAI

Monitoring

Availability monitoring

- node down/drained triggered (when it happens) by strigger via two new systemd services
- fans/GPUfans hourly by crontab script

Performance Monitoring

- Cacti detailed graphs https://ifi-smokeping.uibk.ac.at/cacti [https://ifi-smokeping.uibk.ac.at/cacti]
- Smokeping HERE [https://ifi-smokeping.uibk.ac.at/smokeping/?target=IFIcluster]

Contact

Contact ifi-sysadmin@informatik.uibk.ac.at if you have problems or further questions. For administration purposes there is also the internal documentation.

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